# Tool-Augmented Language Model Agents for Explainable Mine Safety Risk Analysis: A Study Using U.S. Mine Safety and Health Administration Data

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**Abstract**

Mining remains one of the more hazardous industrial occupations, and safety regulators such as the United States Mine Safety and Health Administration (MSHA) collect large volumes of accident and injury data every year, combining structured fields with free text narratives written by mine operators. Prior work on this dataset has treated it as a supervised learning problem: predicting injury severity or days away from work using logistic regression, decision trees, or neural networks trained on structured fields and, in some cases, the narrative text. These models produce a prediction but not an explanation a safety officer can act on, and none of them let a user ask a follow-up question in plain language. Separately, a recent line of work on tool-augmented large language model agents has shown that an LLM can plan, call external tools, and reason over heterogeneous operational data in domains such as oil and gas drilling, but this pattern has not yet been applied to occupational mine safety data, and none of the agentic systems in adjacent domains report a human evaluation of the explanations they produce. This paper proposes and evaluates a tool-augmented LLM agent that answers natural language safety questions by combining a structured risk classifier, a trend analysis tool, and a retrieval tool over historical injury narratives from the MSHA Accident, Injury, and Illness dataset. The system is compared against a plain classifier baseline and a retrieval-augmented generation baseline on answer accuracy, tool selection correctness, and latency. On a 60-question benchmark (20 classification, 20 trend, 20 case-grounded), the primary evaluation uses a live LLM (Llama 3.3 70B via Groq) that selects tools from natural language. The tool-augmented agent achieves **38.3% overall accuracy** versus **30.0%** for the classifier baseline and **28.3%** for the RAG baseline, with 60% tool selection correctness. All systems attempt every question. A deterministic offline router (no LLM reasoning) reaches **93.3%** accuracy as a zero-cost ablation ceiling only. The structured classifier tool alone achieves 0.574 accuracy and 0.562 macro F1 on a 48,128-record holdout over ten injury severity classes. Explanation quality assessment using the Explanation Satisfaction Scale of Hoffman et al. [2] is prepared for mining engineering faculty and senior students; participant data collection is ongoing.

**Keywords:** mine safety; MSHA; tool-augmented LLM; agentic AI; explainable AI; occupational safety; retrieval-augmented generation; human evaluation.

## 1 Introduction

Mining safety has improved substantially over the past several decades through regulation, mechanization, and safety culture, but injuries and fatalities have not been eliminated, and mine

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operators and safety officers still spend considerable effort reviewing historical incident records by hand or through simple dashboards when trying to understand what drives risk at a given site. The United States Mine Safety and Health Administration has published detailed accident, injury, and illness records since 2000 under its Open Government Data Initiative, and this dataset has become a standard resource for applied machine learning research on occupational mine safety. Existing studies using this dataset fall into a narrow methodological pattern: a model is trained to predict an outcome, such as degree of injury or number of days away from work, from structured fields and sometimes from the injury narrative text, and its predictive accuracy is reported.

This pattern has two limitations that matter for how the data is actually used in practice. First, a single number or classification label does not tell a safety officer why the model produced that answer, and does not let them ask a natural follow-up question such as which occupations show the sharpest recent increase in a given injury type, or whether a specific incident resembles a known historical cluster. Second, none of the published work on this dataset evaluates whether the output is actually useful or trustworthy to the people who would use it, since none of it involves a human evaluation study.

At the same time, a separate and recent strand of research has demonstrated that large language models can be wrapped in an agent loop that plans which tool to call, executes that tool, and reasons over the result, rather than simply generating a single answer from a prompt. This pattern, sometimes called tool use or tool augmentation, was formalized in general purpose form by Yao et al. [7] and Schick et al. [5], and has recently been applied to a real industrial operations problem by Lu [4], whose TADI system orchestrates twelve domain-specific tools over drilling reports and sensor data from an oil field, explicitly avoiding heavyweight agent frameworks in favor of a small, auditable orchestration loop built directly on a large language model provider's function calling interface. As far as we can determine from a review of the mining, safety, and agentic AI literature, this pattern has not yet been applied to mine safety data, and the general MLLM agent literature that does touch on mining, notably the MineAgent system of Wang et al. [10], addresses mineral exploration from remote sensing imagery rather than occupational safety or operational decision support.

This paper makes three contributions. First, it introduces a tool-augmented LLM agent architecture for reasoning over the MSHA accident and injury dataset, combining a structured classifier, a statistical trend tool, and a retrieval tool over incident narratives, following the transparent, framework-free orchestration design demonstrated by Lu [4] in a different industrial domain. Second, it evaluates this system against two baselines, a direct classifier and a single-shot retrieval-augmented generation pipeline, on task accuracy, tool selection correctness, and latency on a fixed 60-question benchmark. Third, it provides a human evaluation protocol based on the Explanation Satisfaction Scale of Hoffman et al. [2] for mining engineering faculty and senior students; participant data collection is in progress.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and preprocessing. Section 4 describes the proposed architecture. Section 5 describes the experimental design. Section 6 reports results. Section 7 discusses contributions and limitations. Implementation steps are documented in the repository reproduction guide.

## 2 Related Work

### 2.1 Predictive modeling of MSHA accident and injury data

The MSHA Accident, Injury, and Illness dataset, collected under 30 CFR Part 50 and published through MSHA's Open Government Data Initiative [6], has been used in several prior studies.

Amoako et al. [1] applied multiclass logistic regression to a ten-year injury dataset to identify risk factors associated with different injury classes. Yedla et al. [8] used roughly 228,000 records with fifty variables to train deep neural networks predicting both injury outcome and days away from work, comparing against logistic regression and exploring synthetic data augmentation on narrative text. Yedla [9] applied decision trees, random forests, and artificial neural networks to accidents reported between 2000 and 2018. Across all three studies, the narrative text field, when used at all, is treated as an additional input feature for a supervised model rather than as something a user can query directly, and none includes a human evaluation of how useful or interpretable practitioners find the output.

### 2.2 Computer vision, digital twins, and multimodal AI in mining safety

A parallel body of applied work uses computer vision to detect hazards, personal protective equipment violations, and unsafe proximity between workers and equipment from camera feeds. Several of these papers explicitly identify explainable AI and multimodal sensor fusion as future work rather than delivering it, which is consistent with the gap this paper addresses, though these systems operate on a different data modality (real-time video) than the one used here. Digital twin approaches to mining operations have been the subject of recent systematic reviews; these indicate that the digital twin literature is comparatively mature and would need a narrow, specific technical angle to contribute something new, which is part of why this paper does not pursue that direction.

### 2.3 Tool-augmented and agentic LLM systems

The general capability of large language models to interleave reasoning with calls to external tools was formalized by Yao et al. [7] in the ReAct framework and extended by Schick et al. [5] in Toolformer. Lu [4] applied this pattern to structured drilling records and free text daily drilling reports from the Equinor Volve field, orchestrating twelve domain-specific tools through direct function calling. Wang et al. [10] apply a multimodal LLM agent to mineral exploration from remote sensing imagery, which shares a domain label with the present work but addresses a different task rather than operational or safety decision support. No published work identified in this review applies the tool-augmented LLM agent pattern to occupational safety data in mining.

### 2.4 Explainability and trust evaluation

Hoffman et al. [2] developed validated scales for explainable AI research covering explanation goodness, user satisfaction, curiosity, and trust, including the Explanation Satisfaction Scale. Trust in automation more broadly has a long history in human factors research [3], whose framework for appropriate reliance underlies much of the more recent human-AI collaboration literature this project draws on for study design.

## 3 Data

The primary dataset is the MSHA Accident, Injury, and Illness dataset, available through MSHA's Open Government Data Initiative [6], containing all accidents, injuries, and illnesses reported by mine operators and contractors in the United States since January 1, 2000, drawn from MSHA Form 7000-1. Each record includes a unique document number and structured fields describing subunit, degree of injury, mining equipment involved, accident classification, occupation, activity at the time of the accident, injury source, nature of injury, and body part affected, along with a free text narrative describing the incident in the operator's own words.

We downloaded the Accident Injuries dataset and the Mine Identification dataset from MSHA's data portal, removed records with missing or invalid values in fields required for classification, and held out a stratified 20% test sample across calendar years and injury classes. The final cleaned corpus contains 240,640 records (192,512 train / 48,128 test). Narrative text was embedded with `sentence-transformers/all-MiniLM-L6-v2` and indexed in ChromaDB for semantic retrieval. A benchmark of 60 natural language questions, written before system evaluation, spans classification, trend, and case-grounded categories (Section 5). Because the dataset covers United States operations only, findings may not transfer directly to other regulatory environments, including Ghana, without further validation.

## 4 System Architecture

The system follows the general orchestration pattern demonstrated by Lu [4] for drilling operations, adapted here for occupational safety reasoning rather than drilling engineering, and built without a heavyweight agent framework so that every step in the reasoning process is visible and can be audited.

**Data layer.** Structured MSHA fields are loaded into a relational analytical store. Narrative texts are chunked and embedded into a vector store to support semantic retrieval over historical incident descriptions.

**Tool layer.** Three tools are implemented as deterministic, testable functions:

- A *risk classification tool*, which reproduces and extends the supervised modeling approach of prior MSHA studies (a random forest baseline) to predict injury severity class for a specified set of conditions.
- A *trend analysis tool*, which computes rates, counts, and changes over time for a specified injury type, occupation, equipment category, or mine site.
- A *narrative retrieval tool*, which performs semantic search over historical incident narratives and returns the most relevant matching incidents with their structured metadata.

**Orchestrator.** A large language model receives the user's question, a system prompt describing the three tools, and access to the tools through the provider's native function calling interface. The model selects which tool or sequence of tools to call, receives the tool output, and produces a natural language answer citing which tools were used.

**Baselines for comparison.** Two baselines isolate the value of tool augmentation: a direct classifier baseline, which answers only classification-style questions; and a single-shot retrieval-augmented generation baseline, which retrieves relevant narrative chunks and passes them to the language model in a single prompt without tool orchestration or access to the trend or classification tools.

## 5 Experimental Design

**Research questions.** Does tool-augmented orchestration improve answer accuracy over a direct classifier and over a single-shot retrieval-augmented generation baseline, for natural language safety questions posed against the MSHA dataset? Does the system correctly select the tool or tools appropriate to a given question? What is the latency per query, and how does this scale with question complexity? Do mining engineering domain experts rate the system's explanations as satisfactory and trustworthy using a validated instrument?

**Benchmark construction.** Sixty natural language questions were authored in three equal categories: classification-style questions with explicit MSHA field codes, trend and comparison questions requiring aggregation, and case-grounded questions requiring retrieval of similar historical incidents. Reference answers were derived from cleaned data and tool outputs before any benchmark run.

**Quantitative metrics.** Task accuracy (proportion correct per category and overall), tool selection correctness (invoked tools vs. expected tools per question), and latency (wall-clock time per query).

**Human evaluation.** Mining engineering faculty and senior students at the University of Mines and Technology will rate blinded question–answer pairs using the Explanation Satisfaction Scale [2]. Target sample: 10–20 participants; results will be reported descriptively as an exploratory study.

**Failure case analysis.** Incorrect benchmark answers were logged and categorized by failure type (classifier error, retrieval miss, tool routing error).

## 6 Results

All quantitative results were produced from the open source implementation. Benchmark questions and reference answers were fixed before evaluation. The primary agent evaluation used deterministic tool routing without LLM inference, isolating tool correctness at zero API cost. Supplementary live LLM runs (local Ollama) are reported in Section 6.6.

### 6.1 Data and classifier tool

After cleaning, **240,640** accident records remained (273,614 raw; exclusions documented in the reproduction guide). An 80/20 stratified split yielded 192,512 training and 48,128 test records across ten `DEGREE_INJURY_CD` classes (codes 01–10).

The random forest classifier (`InjuryRiskClassifier`, 100 trees, balanced class weights) achieved on the stratified holdout:

**Table 1:** Classifier holdout metrics on 48,128 test records.

| Metric | Value |
|---|---|
| Accuracy | 0.574 |
| Macro F1 | 0.562 |
| Weighted F1 | 0.565 |
| Fatality (01) recall | 0.538 |

Out-of-time evaluation (train 2000–2020, test 2021+) yielded accuracy 0.553 and macro F1 0.559. These figures are comparable in spirit to prior MSHA supervised modeling at similar data scale [8] but are not directly comparable across different targets and feature sets.

### 6.2 Primary benchmark (Groq llama-3.3-70b)

The primary reported result uses Groq `llama-3.3-70b-versatile`. The LLM plans tool calls from natural language. All three systems attempt all 60 questions.

Mean latency: agent 126 s, classifier 0.19 s, RAG 256 s per question.

Table 2: Primary live LLM benchmark (Groq).

| System | Class. | Trend | Case | Overall | Tool sel. |
|---|---|---|---|---|---|
| Tool-augmented agent | 85.0% | 5.0% | 25.0% | **38.3%** | 60.0% |
| Classifier baseline | 90.0% | 0.0% | 0.0% | 30.0% | 100% |
| RAG baseline | 0.0% | 0.0% | 85.0% | 28.3% | 98.3% |

### 6.3 Offline routing ablation (deterministic, no LLM)

A separate run uses category metadata to route questions without LLM inference. This measures a zero-cost routing ceiling, not live agentic reasoning.

Table 3: Offline routing ablation ($n = 20$ per category).

| System | Classification | Trend | Case-grounded | Overall |
|---|---|---|---|---|
| Offline router | 90.0% | **100.0%** | 90.0% | **93.3%** |
| Classifier baseline | 90.0% | 0.0% | 0.0% | 30.0% |
| RAG baseline (no LLM) | 0.0% | 0.0% | 90.0% | 30.0% |

The 30% overall scores for single-tool baselines reflect incorrect answers on out-of-strength questions after attempting every question, not category blocking. Mean latency: offline router 0.31 s, classifier 0.22 s, RAG 0.03 s.

### 6.4 Failure analysis (offline router)

Four agent failures occurred:

1. **Two classification errors (CLS-03, CLS-14):** classifier predicted degree code 03 instead of the reference code.
2. **Two retrieval errors (CASE-14, CASE-15):** semantically similar narratives were retrieved but the reference document was not in the top five results.

Baseline failures on out-of-domain question types are expected when a single tool is used for every question.

### 6.5 Human evaluation

Materials for the Hoffman et al. [2] Explanation Satisfaction Scale are prepared. Participant ratings are not yet included; data collection at the University of Mines and Technology may proceed now that the primary Groq benchmark is complete.

### 6.6 Live LLM evaluation (Ollama, supplementary)

A supplementary run used Ollama with `qwen2.5:7b` before baseline correction (agent 53.3% overall, 88.3% tool selection). That run is not directly comparable to the corrected Groq comparison above.

## 7 Contributions and Limitations

This paper makes three contributions, now supported by implemented code and measured results. First, it introduces a tool-augmented LLM agent architecture for reasoning over the MSHA accident and injury dataset. Second, on a fixed 60-question benchmark where every system attempts every question, the live Groq agent achieves 38.3% overall accuracy versus 30.0% for the classifier baseline and 28.3% for the RAG baseline. Third, it prepares the first human evaluation protocol for explanation quality in mine safety AI using the Explanation Satisfaction Scale of Hoffman et al. [2]; participant data collection remains future work.

Limitations stated plainly:

- **Baseline comparison correction:** An earlier draft restricted each baseline to its home category; the corrected comparison attempts all 60 questions with all systems.
- **Offline routing ablation:** The 93.3% offline router result uses category metadata, not live LLM reasoning.
- **Live LLM weakness:** Primary Groq agent accuracy (38.3%) is far below the offline ablation; trend questions scored 5%.
- **Geographic scope:** U.S. MSHA data only; findings may not transfer to Ghana or other regulatory environments without validation.
- **Classifier weakness:** Macro F1 0.562; several severity classes (04, 09, 10) have low recall.
- **Retrieval:** Semantic search sometimes returns plausible but incorrect reference documents (two offline router failures).
- **Human evaluation:** Protocol prepared; participant data not yet collected.

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